ARJUNA JEE 2027

Lecture -04 | Mathematics

PERMUTATIONS AND COMBINATIONS

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Topics *to be covered*

A Formula for P & C

B Problems on Combination(Selection)

C Problems on Permutation (Arrangement)

D

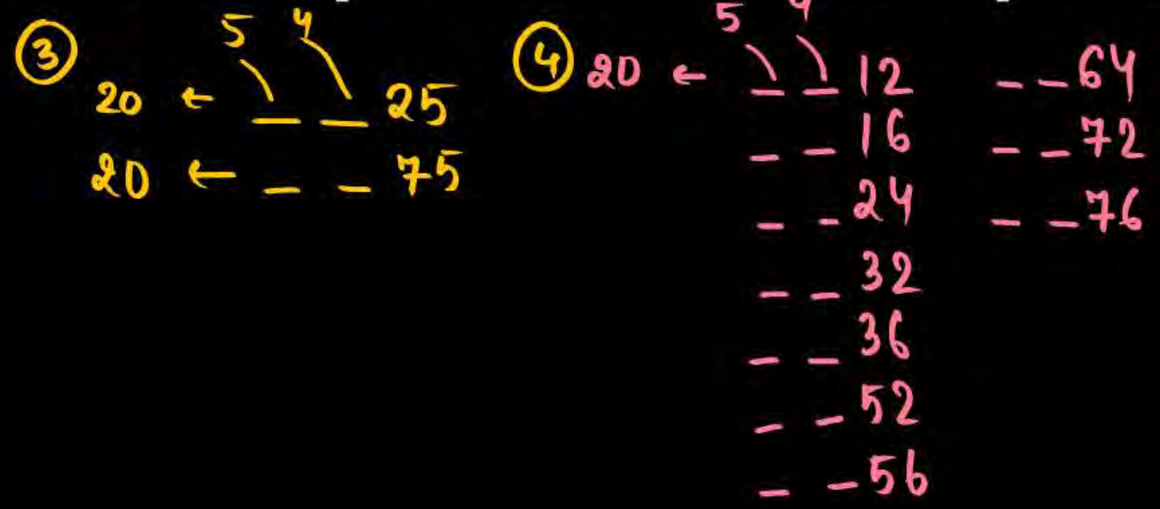


H.W. Discussion

Question

Using the digits 1, 2, 3, 4, 5, 6, 7, a number of 4 different digits is formed. Find

C_1	C_2
A (1) How many number are formed?	(A) 840
C (2) How many numbers are exactly divisible by 2?	(B) 200
D (3) How many numbers are exactly divisible by 25?	(C) 360
B (4) How many of these are exactly divisible by 4?	(D) 40
	(E) 180



Question [JEE Mains-2023]

The number of five-digit numbers, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to



A 132

B 120

C 72

D 96

$5 _ _ _ 0$ $\}$ 24 ✓

$7 _ _ _ \square$ $\}$ 48. ✓

$9 _ _ _ \square$ $\}$ 48. ✓

$4 \times 3 \times 2 \times 2$ opt. (0/5)

Question [JEE Mains-2022]

Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits 1, 2, 3, 4, 5 and 6 without repetition of digits. Then the total number of such numbers is _____ 30 _____.

1 _ 2 4
1 _ 3 2
1 _ 3 6

Done!

Question

Find the number of words that can be formed using all the letters of the word 'MIRACLE' such that the vowels occupy only the odd positions.

A, I, E

M, R, C, L

	1	2	3	4	5	6	7
	*		*		*		*
I	V	—	V	—	V	—	—
II	V	—	—	—	V	—	V
III	V	—	V	—	—	—	V
IV	—	—	V	—	V	—	V

I → $^{3}_{0}!\times{}^{4}_{0}! = 144$

II → "

III → "

IV → "

$fnf = 144 \times 4$

$= 576.$

- A. 576
- B. 144
- C. 288
- D. 720

FORMULA's FOR P & C

Theorem-01 :

Number of ways of arranging n different things in a line = $n!$

Theorem-02 :

Permut.

Number of ways of arranging r things taken out of n-different things in a line = $^{n}P_{r} = \frac{n!}{(n-r)!}$

Theorem-03 :

Number of ways of selecting r things out of n-different things = $^{n}C_{r} = \frac{n!}{r!(n-r)!}$

Comb.

RAPID FIRE ROUND

Answer the following Questions:

1. Value of ${}^{10}C_3$ = 120.
2. If ${}^{12}C_x = {}^{12}C_7$ then possible value(s) of x = 7, 5.
3. Value of ${}^{7}P_3$ = 210.
4. (True/False) : ${}^{n}C_0 = {}^{n}C_n = 1$

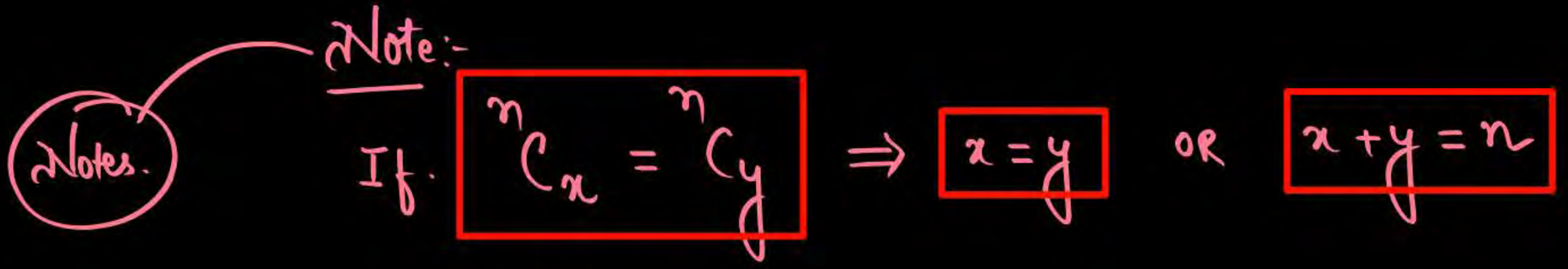
(i) ${}^{10}C_3 = \frac{10!}{7!\,3!} = \frac{10.9.8.7.....1}{7!.3.2.1} = 120.$

(2) ${}^{12}C_x = {}^{12}C_7 \rightarrow x = 7$

$x + 7 = 12$

$x = 5$

${}^{7}P_3 = \frac{7!}{4!} = \frac{7.6.5.4!}{4!} = 210.$


Note:-
Notes.
If. ${}^{n}C_{x} = {}^{n}C_{y}$
$\Rightarrow$ $x = y$
OR
$x + y = n$

Question

If ${}^{2n}P_4 = 396 \times {}^{n}P_2$, then the value of n, is

A 4

B 5

C 6

D 7

HW.
(80% DIBY)

RAPID FIRE QUESTION

Total = (11) stds.

In a class there are six boys & five girls, then:

(i) Number of ways of selecting 5 students. → $^{11}C_5$ ✓

(ii) Number of ways of selecting 3 girls & 2 boys.

(iii) Number of ways of selecting 5 students with atleast 2 boys.

(iv) Number of ways of selecting 5 students with atmost 2 girls.

(ii) 5G & 6B.

$^{5}C_3 \times {}^{6}C_2$

(iii)

fnf. = $^{6}C_2 \times {}^{5}C_3$ (2B, 3G) + $^{6}C_3 \times {}^{5}C_2$ (3B, 2G) + $^{6}C_4 \times {}^{5}C_1$ (4B 1G) + $^{6}C_5$ (5B).

(iv) 3B 2G + 4B 1G + 5B 0G

RAPID FIRE QUESTION

In a class there are six Boys & five Girls, then:

(i) Number of ways of arranging 3 Boys & 2 Girls in a line such that end positions occupied by girls.

(ii) Number of ways of arranging 3 Boys & 2 Girls in a line such that end positions occupied by boys.

(iii) Number of ways of arranging 3 Boys & 2 Girls in a line such that boys and girls are alternate.

(i) Select. Arrang.

$${}^6C_3 \times {}^5C_2 \times \underline{\ominus}\ \underline{\ \ }\ \underline{\ \ }\ \underline{\ \ }\ \underline{\ominus}$$

$2 \times 3 \times 2 \times 1 \times 1$

$B_1B_2B_3G_1G_2$

$= {}^6C_3 \times {}^5C_2 \times 12$

(ii) Select. Arrang.

$${}^6C_3 \times {}^5C_2 \times 36$$

$\ominus\ \underline{\ \ }\ \underline{\ \ }\ \underline{\ \ }\ \underline{\ \ }\ \ominus$

$3 \times 3 \times 2 \times 1 \times 2$

(iii) ${}^6C_3 \times {}^5C_2$

B G B G B

$3 \times 2 \times 2 \times 1 \times 1$

BGBGB ✓

GBGBB ✗

$\times 12$

11 stds.

6B, 5G

no. of ways of arranging 3B, 2G in a line:

Selecⁿ. × arrang.

$${}^{6}C_{3} \times {}^{5}C_{2} \times 5!_{0}$$ ✓

$${}^{11}P_{5} = \frac{11!_{0}}{6!_{0}}$$

X

RESRTICTED SELECTION EXAMPLE

5 boys are selected out of 9 boys such that :

(i) Two particular boys will not serves together.

(ii) Two specific boys will only serves together.

already fixed

9 Boys → A, B

9 Boys → 7 Boys.

(i) $^9C_5 - \left(^7C_3\right)$ (A ✓, B ✓)

OR

equal

(i) $= \left(^7C_4 + {}^7C_4 + {}^7C_5\right)$

(A ✓, B ✗) (A ✗, B ✓) (A ✗, B ✗)

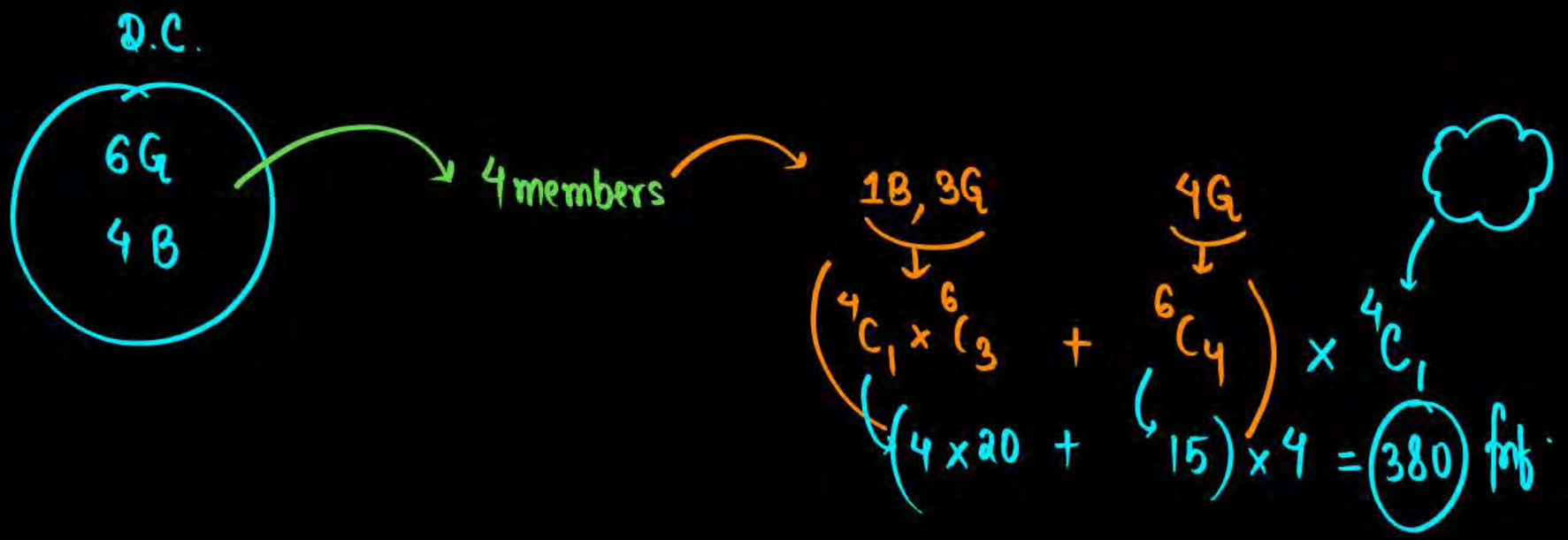
(ii) $= \left(^7C_3 + {}^7C_5\right)$

(A ✓, B ✓) (A ✗, B ✗)

FORMATION OF TEAM / COMMITTEE

Question [JEE (Adv.)-2016, Paper-1]

A debate club consists of 6 girls and 4 boys. A team of 4 members is to be selected from this club including the selection of a captain (from among these 4 members) for the team. If the team has to included at most one boy, then the number of ways of selecting the them is_____.



Question [JEE Mains-2019]

A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then



A $m = n = 68$

B $m + n = 68$

C $m = n = 78$

D $n = m - 8$

Question [JEE (Adv.)-2018, Paper-2]

In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5.

(i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.

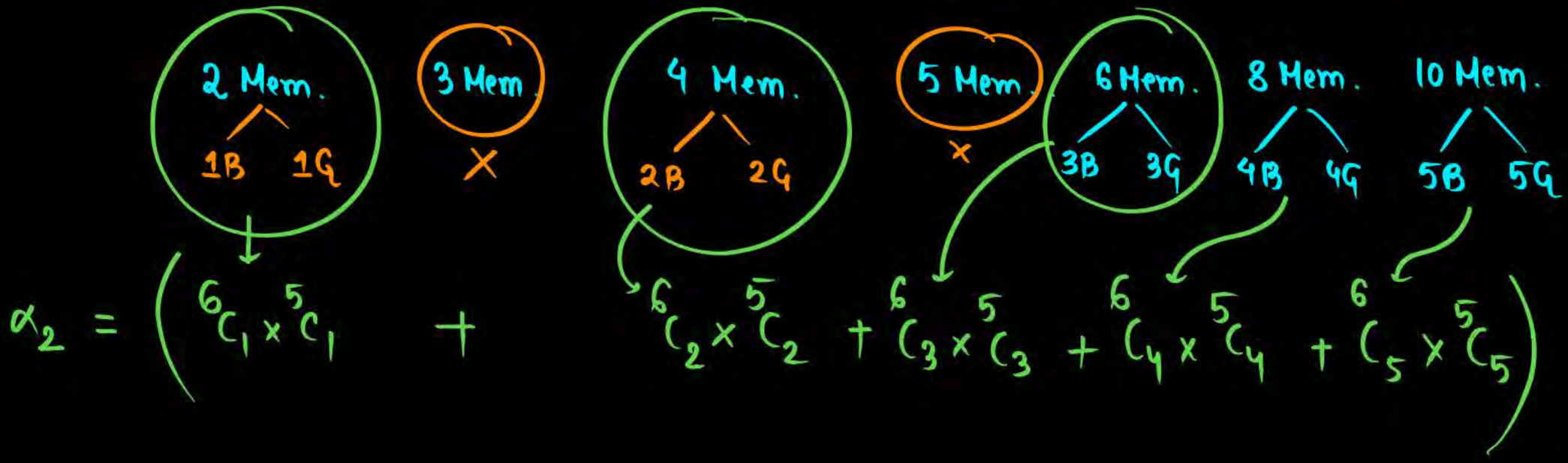
(3B 2G)

$$\alpha_1 = {}^6C_3 \times {}^5C_2$$

Question [JEE (Adv.)-2018, Paper-2]

In a high school, a committee has to be formed from a group of 6 boys M_1, M_2, M_3, M_4, M_5, M_6 and 5 girls G_1, G_2, G_3, G_4, G_5.

(ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members and an equal number of boys and girls.



Question [JEE (Adv.)-2018, Paper-2]

In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5.

(iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.

$\alpha_3 =$ $\underbrace{2G\,3B}$ $+$ $\underbrace{3G\,2B}$ $+$ $\underbrace{4G\,1B}$ $+$ $\underbrace{5G}$

Question [JEE (Adv.)-2018, Paper-2]

In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5.

(iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.



4 Members. | at least 2 Girls. | M_1 & G_1 are not together.

2G 2B | 3G 1B | 4G

$+ {}^5C_4$

4 Members. | at least 2 Girls. | M_1 & G_1 are not together.

$M_1\ M_2\ M_3\ M_4\ M_5\ M_6$

$G_1\ G_2\ G_3\ G_4\ G_5$.

2G 2B + 3G 1B + 4G

$^5C_2\ ^6C_2 - (^5C_1 \times {}^4C_1)$

↑ M_1G_1 are together

$10 \times 15 - 5 \times 4$

$150 - 20$.

M_1 ✓ G_1 ✗ | M_1 ✗ G_1 ✓ | M_1 ✗ G_1 ✗ + $^5C_4$

$^4C_3 + {}^4C_2 \times {}^5C_1 + {}^5C_1 \times {}^4C_3$

$4 + 6 \times 5 + 5 \times 4 = 54$

$\alpha_4 = 130 + 54 + 5$

OR

$^5C_3 \times {}^6C_1 - (^{M_1G_1 \text{ are together}}_{\ \ {}^4C_2}) = 60 - 6 = 54$

Question [JEE (Adv.)-2018, Paper-2]

In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5.

(iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.



4 Members. | at least 2 Girls. | M_1 & G_1 are not together.

Question

The kinder garden teacher has 25 kids in her class. She takes 5 of them at a time, to zoological garden as often as she can, without taking the same 5 kids more than once. Then the number of visits, the teacher makes to the garden exceeds that of a kid by



(M)

$\{a, b, c, d, e\}$

$10 = {}^{5}C_{3}$

abc
abd
abe
acd
ace
ade
bcd
bce
bde
cde

a _ _ _

$^{4}C_{2} = 6$

Example

A dice is rolled 3 times, top-most face is considered as outcome then find the number of outcome such that :

(i) 5 appears atleast once.

(ii) All 3 outcomes are different.

(iii) Exactly 2 outcomes are same.

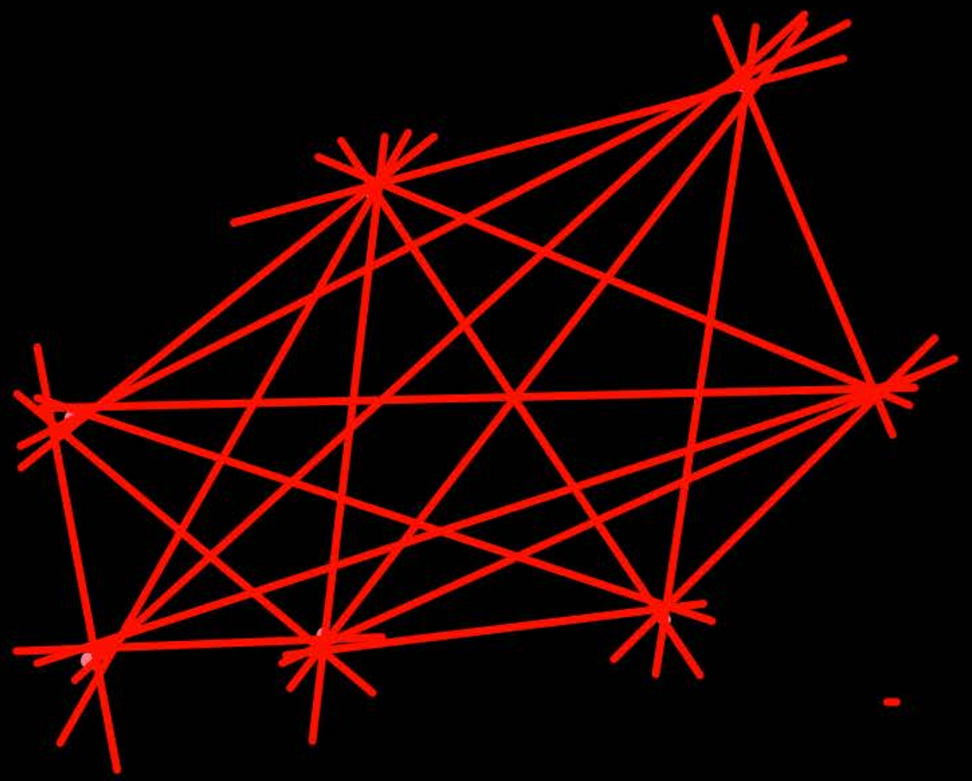


A B C D E F G

$6+5+4+3+2+1 = 21$

OR

$= {}^{7}C_{2} \times 1 = \frac{7!}{2!\,5!} = \frac{7 \cdot 6 \cdot 5!}{2 \times 5!} = 21$

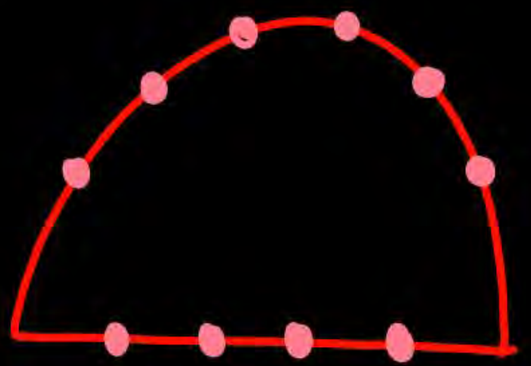


$^{7}C_{2} \times 1 = 21.$

GEOMETRICAL USES OF ${}^{n}C_{r}$

Example

Total 10 points are there, out of which 4 are collinear, then find total number of different (i) straight line & (ii) triangles that can be formed by joining these points.



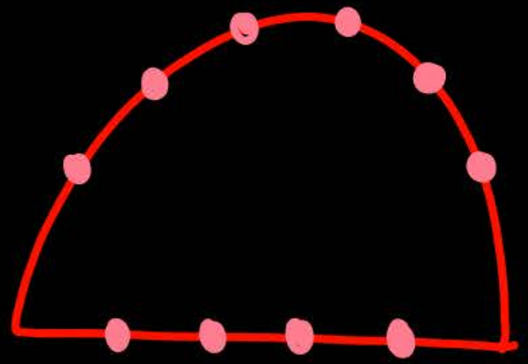
(i) St. line

M-I: ${}^{6}C_{2} \times 1 + 1 + {}^{6}C_{1} \times {}^{4}C_{1} \times 1$

$= 15 + 1 + 24 = 40$

M-II: ${}^{10}C_{2} \times 1 - {}^{4}C_{2} \times 1 + 1 = 40$

$45 - 6 + 1$



(ii) Triangles :-

M-I :-

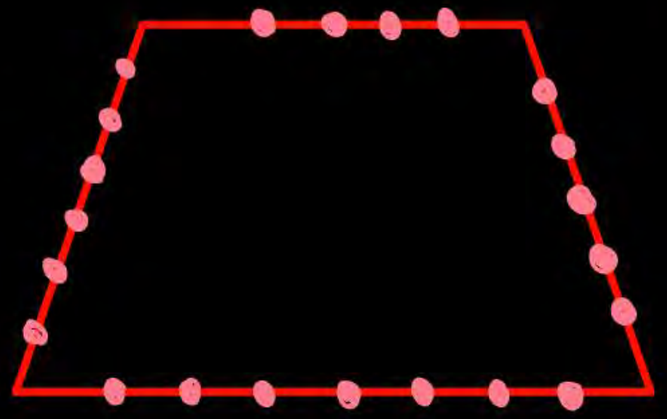
$$^{6}C_{3} + {}^{6}C_{2} \times {}^{4}C_{1} + {}^{6}C_{1} \times {}^{4}C_{2} = 116.$$

(20, 15 × 4, 6 × 6)

M-II :-

$$^{10}C_{3} - {}^{4}C_{3} = 120 - 4 = 116.$$

HW.



① No. of st. lines.

② No. of Δ's.

Question Practice

Challenger Question [JEE Mains-2022]

The number of 7-digits numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is _____.



Question [JEE Mains-2023]

The number of integers, greater than 7000 that can be formed, using the digits 3, 5, 6, 7, 8 without repetition, is.



A 120

B 168

C 220

D 48

Question [JEE Mains-2022]

Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from $\{A, B, C, D, E\}$ or a number from {1, 2, 3, 4, 5} with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from {1, 2, 3, 4, 5} is $\alpha \times 5^6$, then α is equal to

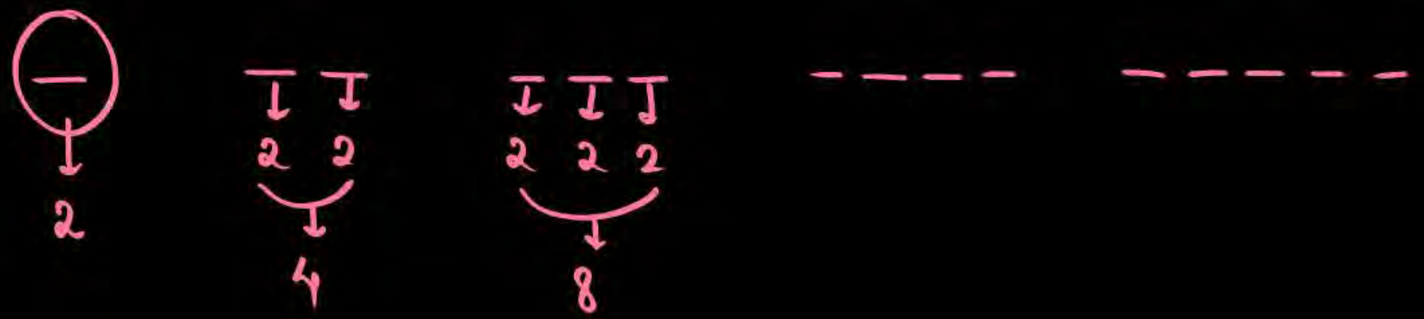


DO IT BY YOURSELF (DIBY)

DIBY-05

How many positive integers less than 10^5 can be written with only the digits 7 and 9 ?

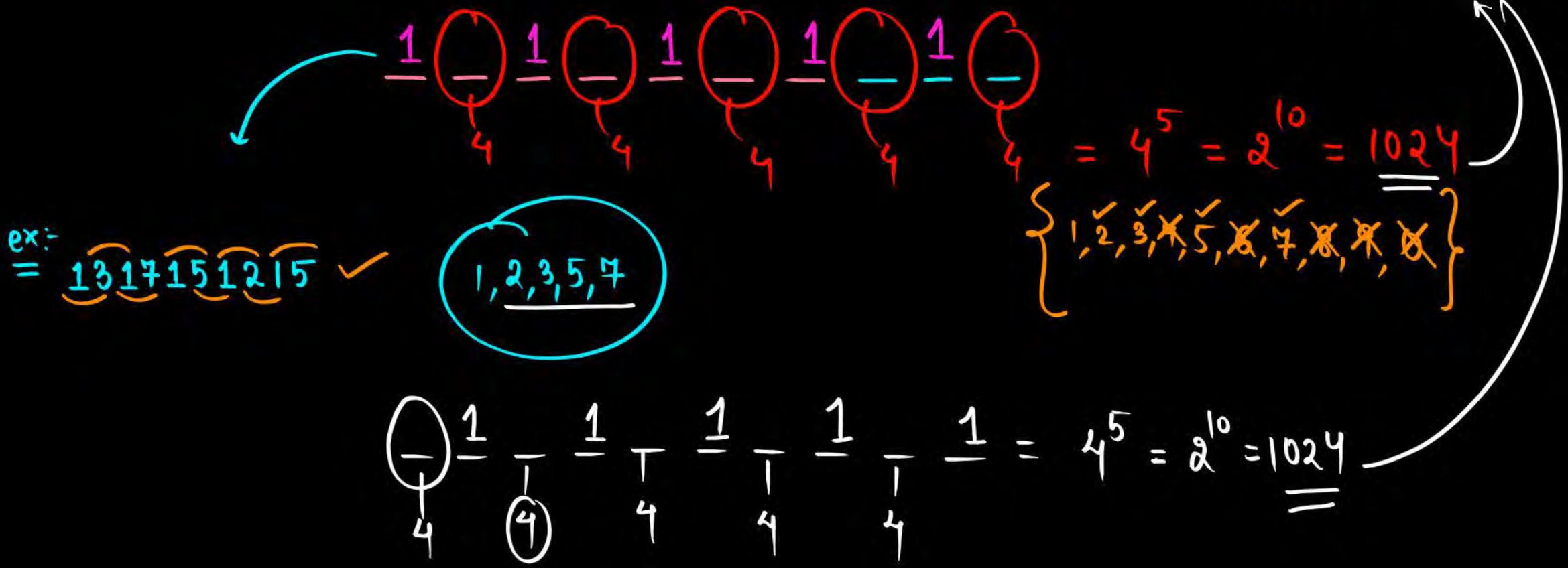
[Ans. : 62]



DIBY-07

Find the number of 10-digit numbers such that the product of any two consecutive digits in the number is a prime number.

[Ans. : 2048]



DIBY-08

Find number of ways in which 7 different colours in a rainbow can be arranged, if green is always in the middle.

[Ans. : 720]

DIBY-09

Find the number of natural numbers between 100 and 1000 such that atleast one of their digits is 7 .

[Ans. : 252]

DIBY-10

If the digits are not allowed to repeat in any numbers formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to

[JEE Mains-2021]

[Ans. : 96]

DIBY-11

Find the number of 4 digit integers in the closed interval [2022, 4482] which can be formed using the digits {0, 2, 3, 4, 6, 7}. [JEE (Adv.)-2022]

[Ans. : 569]

DIBY-12

The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal

[JEE Mains-2024]

[Ans. : 36]

DIBY-13

Total numbers of 3-digit numbers that are divisible by 6 and can be formed by using the digits 1, 2, 3, 4, 5 with repetition, is _____. [JEE Mains-2023]

[Ans. : 16]

DIBY-14

The number of words that can be formed out of the letters of the word ARTICLE so that the vowels occupy even places is

(A) 36 (B) 574 (C) 144 (D) 754

[Ans. : (C)]

DIBY-15

Eleven books consisting of 5 Mathematics, 4 Physics and 2 Chemistry are placed on a shelf. The number of possible ways of arranging them on the assumption that the books of the same subject are all together is

(A) 4!2! (B) 11! (C) 5!4!3!2! (D) None of these

[Ans. : (C)]

DIBY-16

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is

(A) 576 (B) 140 (C) 356 (D) 209

[Ans. : (A)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians